

BRAYAN PISSO

ELECTRONIC ENGINEER

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🔗 <https://github.com/BRAYANPISSO02>

ABOUT ME

Electronics Engineer specialized in Machine Learning and Computer Vision, with a focus on developing end-to-end solutions. Expertise in dataset construction, supervised model training, and deploying solutions using PyTorch and Amazon SageMaker. Experience in image classification projects using convolutional neural networks and in developing automated vector representation generation systems applied to design processes in real-world environments, significantly reducing manual design time. Knowledge of data acquisition systems and wireless communications (WiFi HaLow and MESH Networks) applied to production environments. Focused on process automation through artificial intelligence and real-world problem solving.

ACADEMIC BACKGROUND

Electronic Engineer

Universidad Nacional de Colombia

Aug 2020 – Apr 2026

Software Programming Technician

Servicio Nacional de Aprendizaje (SENA)

Feb 2019 – Nov 2019

M.S. Student in Industrial Automation Engineering

(Research Focus: Machine Learning and Automatic Recognition)

Universidad Nacional de Colombia

Jun 2026 – Present

EXPERIENCE

Automatic Generation of Human Vector Representations Using Deep Learning | Small Business

Feb 2026 – Present

Implemented a computer vision solution to automatically generate SVG vector representations of people from photographs. The project, aimed at a microenterprise specializing in laser cutting and engraving, has a real impact by reducing manual design time by approximately two hours per image. It leverages deep learning models developed in PyTorch, including the Segment Anything Model (SAM) for interactive segmentation tasks. A data engineering pipeline was developed to build a dataset for supervised training of a vision model in Amazon SageMaker, focused on image-to-image translation tasks.

Image Processing Project | Universidad Nacional de Colombia

Aug 2024 – Dec 2024

Developed computer vision pipelines for image classification using convolutional neural networks (CNNs) in TensorFlow/Keras, including data preprocessing, supervised training, dropout-based regularization, and experimental performance evaluation across multiple standard datasets.

Long-Range Telemetry Project | Percepción y Control Inteligente (PCI) Research Group.

Jun 2025 – Dec 2025

Developed and implemented a pilot test at the Universidad Nacional de Colombia, La Nubia campus, of a long-distance data acquisition system for multiple electrical energy meters, using Modbus RTU/TCP and DLMS/COSEM protocols over Wi-Fi HaLow connectivity, with integration into the ThingsBoard IoT monitoring and management platform, designed to operate in self-organized multi-hop mesh network topologies.

Responsible for schematic design and PCB routing in KiCad (2D/3D) as well as electronic board prototyping and the development of structures for manufacturing using CNC machining and laser cutting.

COURSES

GenAI and LLMs on AWS | Coursera (Certified)
Duke University

Data Science & Inteligencia Artificial aplicados a negocios | Coursera (Certified)
University of Palermo

AWS S3 Basic | Coursera (Certified)
Coursera

SKILLS

Languages:

English B2 (CEFR), Native Spanish

Technical Skills

- **Programming languages:** Python, Rust, C, C++ | Git (repository management).
- **Cloud Infrastructure:** AWS (Lambda, SageMaker, Bedrock, S3) and Docker.
- **Modeling and simulation tools:** MATLAB, AutoCAD, Proteus, Multisim, and KiCad.
- **Embedded systems:** STM32, ESP32, BeaglePlay, and Raspberry Pi.
- **CNC machinery operation:** Router and Laser.

Soft Skills

Proactivity and initiative, needs analysis, effective communication, resilience, teamwork, leadership.

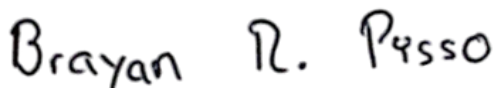
REFERENCES (References contacts available upon request)

GUSTAVO ADOLFO OSORIO LONDOÑO (Ph.D. in Industrial Automation, M.Sc. in Engineering)

Professor, Department of Electrical, Electronic and Communications Engineering
Universidad Nacional de Colombia

LUIS FERNANDO DÍAZ CADAVID (Ph.D., M.Sc.)

Professor, Department of Electrical, Electronic and Communications Engineering
Universidad Nacional de Colombia



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Link to documents

<https://drive.google.com/drive/folders/1gq-mUJ3unNOivBP7oHyEAeMZfHrIhP75?usp=sharing>